Grand Quiz Spring 2021

Subject Code MTH601 lecture 1 to 22

Solved By Riz Mughal

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RIZ MUGHAL (SQA ENGINEER)

MTH601:Grand Quiz

Quiz Start Time: 09:00 AM, 29

Question # 1 of 30 (Start time: 09:00:15 AM, 29 June 2021)

Tot

Which of the following correspond to the practical limitations of available resources while modeling a real life situation into a linear program?

Select the correct option

- (●) Constraints
- ( ) Objective function
- ( ) Linear function
- ( ) Decision variables

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Question # 2 of 30 (Start time: 09:00:43 AM, 29 June 2021)

Which of the following method follows iterative procedure to solve a Linear Programming problem?

Select the correct option

- ○ Graphical
- ● Algebraic (Simplex)

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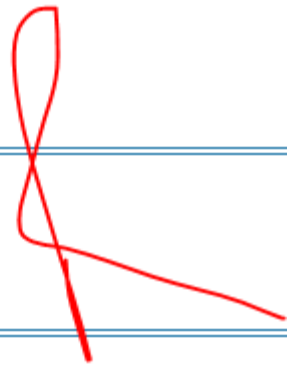
Question # 3 of 30 (Start time: 09:00:57 AM, 29 June 2021)

Solution region of the constraints: '2x+3y>12' and '2x+3y<12' will be the---------.

Select the correct option

- point(6,4)
- straight line:2x+3y=12
- empty set
- all first quadrant



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Question # 4 of 30 (Start time: 09:01:13 AM, 29 June 2021)

In the graph of the Purchasing Model with shortages, the area under the horizontal axis represents ----------

Select the correct option

- set up cost of item
- purchase cost of item
- carrying cost of item
- shortage cost of item

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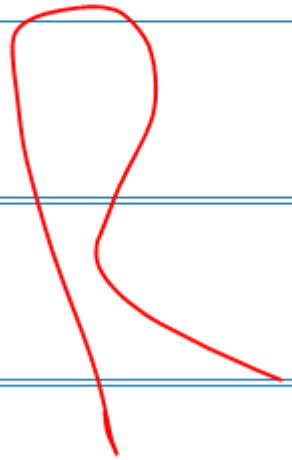
Question # 5 of 30 (Start time: 09:01:27 AM, 29 June 2021)

The straight line associated with the constraint;2x+3y<12,will meet x-axis at a distance of-------- units from origin.

Select the correct option

- 2
- 4
- 6
- 3



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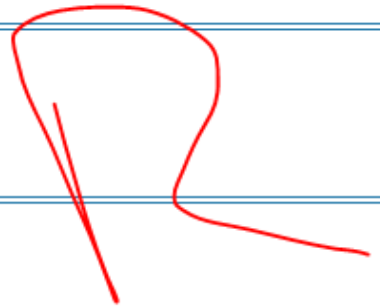
Quiz Start Time: 09:00 AM, 29 Ju

Question # 6 of 30 (Start time: 09:01:51 AM, 29 June 2021) **Total**

Which of the following value is correct for the expected time of an activity having optimistic, pessimistic and most likely times as 4, 8 and 6 days respectively?

Select the correct option

- ○ 6.66 days
- ○ 4.66 days
- ● 6 days
- ○ 9.33 days



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Quiz Start Time: 09:00 AM, 29 June 2

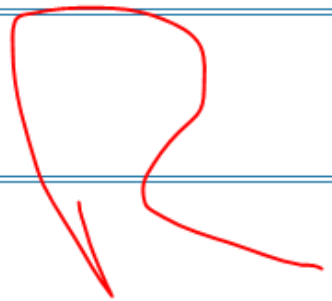
Question # 7 of 30 (Start time: 09:02:10 AM, 29 June 2021)

Total Mark

In a network flow diagram, for an activity (i,j) of duration of three days, if its earliest start time is of two days then which of the following will be its early finish time?

Select the correct option

○	One and half day
○	One day
○	Six days
●	Five days



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Quiz Start Time: 09:00 AM, 29 June 2021

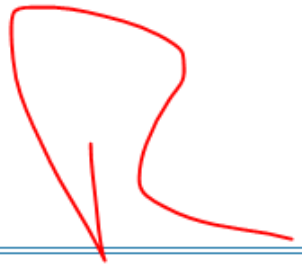
Question # 8 of 30 (Start time: 09:02:26 AM, 29 June 2021)

Total Marks:

Which of the following method would remain impractical to solve a Linear Programming problem when there are more than two decision variable?

Select the correct option

- ○ Algebraic (Simplex)
- ● Graphical



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Quiz Start Time: 09:00 AM, 29 Ju

Question # 9 of 30 (Start time: 09:02:44 AM, 29 June 2021) **Total**

In a development project, if an activity (m,n) of seven days duration, starts late on 5rd day then which of the following will be its latest finish time?

Select the correct option

- 7th day
- 5th or 7th day
- 12th day
- 5th day

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Question # 10 of 30 (Start time: 09:03:00 AM, 29 June 2021)

Which of the fiollowing is an intermediate step to model a linear programming problem?

Select the correct option

- ○ Identify the objective function
- ● Identify the non-negative restrictions and constraints
- ○ Identify the unknown decision variables
- ○ None of these

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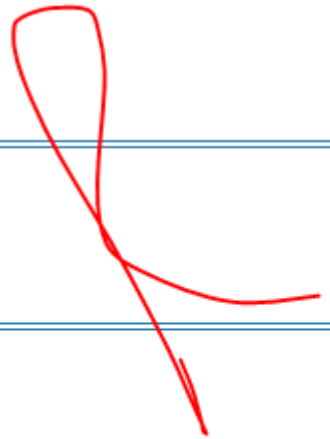
Question # 11 of 30 (Start time: 09:03:16 AM, 29 June 2021)

A/An ---------- is the collection of inter related activities to be performed in a particular sequence to completion.

Select the correct option

- (●) project
- ( ) branch
- ( ) event
- ( ) node



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Quiz Start Time: 09:00 AM, 29 June 2021

Question # 12 of 30 (Start time: 09:03:31 AM, 29 June 2021) **Total Marks:** 1

Which of the following times are directly related to the activity cost that is by reducing the activity duration, the direct cost of the corresponding activity is increased?

Select the correct option

- ○ PERT Times
- ● CPM Times

Click to Save Answer & Move to Next Question

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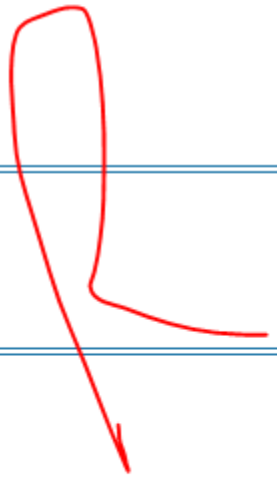
Question # 13 of 30 (Start time: 09:03:49 AM, 29 June 2021)

Which of the following is an operations research (OR) process?

Select the correct option

- ○ Simplex method
- ○ Linear programming
- ● Observation
- ○ Networking



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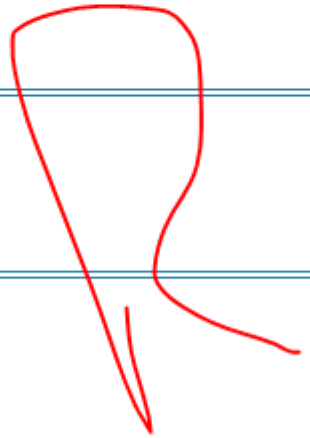
Question # 14 of 30 (Start time: 09:04:04 AM, 29 June 2021)

In the Purchasing Model with Shortages the Cost Function 'C(S,Q)' can also be expressed as function of ----------.

Select the correct option

- (S,D)
- (Q,D)
- (S,t)
- (t,D)



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Quiz Start Time: 09:00 AM, 29 June 20

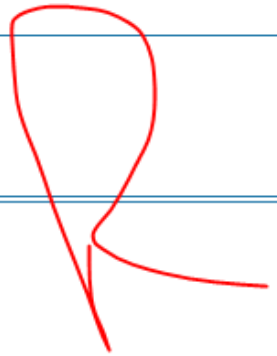
Question # 15 of 30 (Start time: 09:04:21 AM, 29 June 2021)

Total Marks

Which of the following would be the objective of the yield per minute in a chemical process which depends upon the temperature 'x' and pressure 'y'?

Select the correct option

- ○ Minimization
- ○ Inflection
- ● Maximization
- ○ Average



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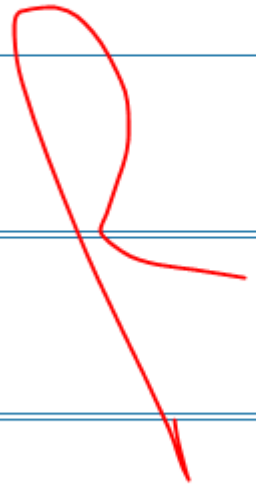
Question # 16 of 30 (Start time: 09:04:38 AM, 29 June 2021)

Which of the following is an operations research (OR) process?

Select the correct option

- ○ Linear programming
- ○ Simplex method
- ○ Networking
- ● Solution



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Quiz Start Time: 09:00 AM, 29 Jun

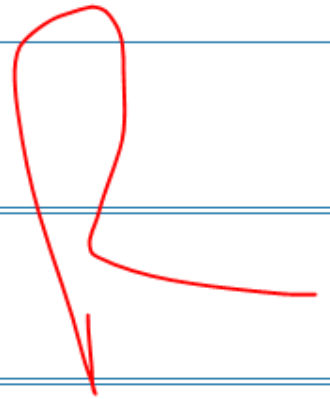
Question # 17 of 30 (Start time: 09:04:54 AM, 29 June 2021)

Total M

In the Dynamic Order Quantity problem if the ratio of Setup and Carrying Costs is '500' and the Demands of 2nd and 3rd months are 50 and 130 respectively then which of the following is true about the 2nd month's requirment?

Select the correct option

- (●) 2nd month demand can be included in 1st month
- ( ) 2nd month demand can be included in 3rd month
- ( ) 2nd month demand can be included in any month of the year
- ( ) 2nd month demand will have to fulfill in 2nd month



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Quiz Start Time: 09:00 AM, 29 June 202

Question # 18 of 30 (Start time: 09:05:09 AM, 29 June 2021) **Total Marks:**

The mathematical technique which is used to solve a wide class of problems such as allocating scarce resources among competitive activities is known as --------.

Select the correct option

- ○ Classical model
- ○ Descriptive model
- ● Linear Programming model
- ○ Mathematical Model

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Question # 19 of 30 (Start time: 09:05:28 AM, 29 June 2021)

Which of the following would be the objective of the cost per unit of producing certain cameras?

Select the correct option

- ○ Average
- ● Minimization
- ○ Inflection
- ○ Maximization

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Question # 20 of 30 (Start time: 09:05:50 AM, 29 June 2021)

In CPM each activity has one deterministic time while in PERT each activity has ----------probabilistic time/times.

Select the correct option

- three
- also one
- two
- no

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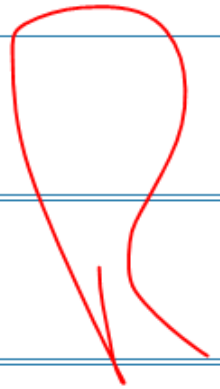
Quiz Start Time: 09:00 AM, 29 June 2021

Question # 21 of 30 (Start time: 09:06:03 AM, 29 June 2021) **Total Marks:** 1

In the phase of Early Start and Early Finish, to find the Critical Path in a network flow diagram, the computations are proceeded from ---------- to the final event.

Select the correct option

- (●) left to right
- ( ) bottom to top
- ( ) top to bottom
- ( ) right to left



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Quiz Start Time: 09:00 AM.

Question # 22 of 30 (Start time: 09:06:19 AM, 29 June 2021)

In a development project, if an activity (i,j) of six days duration, starts late on 3rd day then which of the following will be its latest finish time?

Select the correct option

- ○ 18th day
- ● 9th day
- ○ 3rd day
- ○ 2nd day

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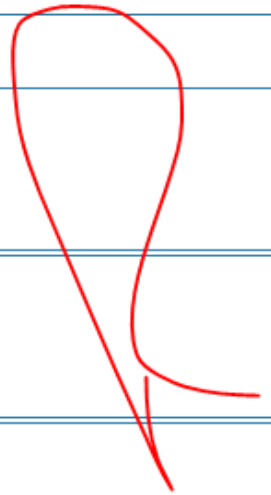
Quiz Start Time: 09:00 AM, 29 June 2(

Question # 23 of 30 (Start time: 09:06:33 AM, 29 June 2021) **Total Mark**

If the annul demand of a product is 10000 items, and its set up and inventory costs are 200 and 100 respectively then its economic order quantity is ---- -----provided that shortage is fulfilled INSTANTANEOUSLY.

Select the correct option

- (●) 200 units
- ( ) 20000 units
- ( ) 40000 units
- ( ) 100 units



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Question # 24 of 30 (Start time: 09:06:49 AM, 29 June 2021)

For a LP problem say;Max:z=x+y,under the constraints x,y>=0, the feasible region would be--------.

Select the correct option

- (●) all the first quadrant
- ( ) all xy-plane
- ( ) empty
- ( ) point(0,0)

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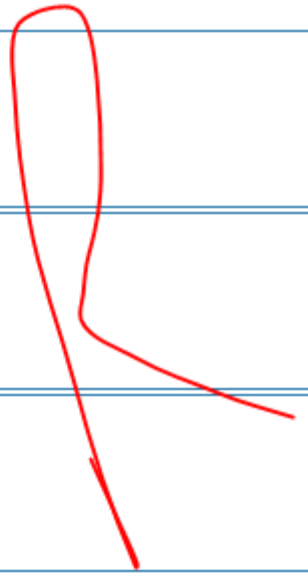
Question # 25 of 30 (Start time: 09:07:06 AM, 29 June 2021)

By which of the following method, any complex linear programming problem can be handled?

Select the correct option

- ○ Degenerate method
- ○ Graphical method
- ● Simplex method
- ○ Dual method



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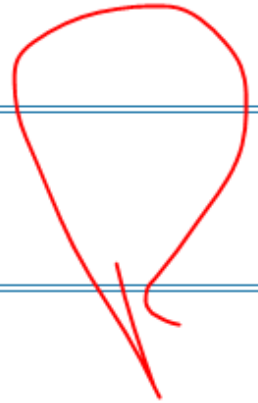
Quiz Start Time: 09:00 AM, 2

Question # 26 of 30 (Start time: 09:07:22 AM, 29 June 2021)

In the phase of Early Start and Early Finish, to find the Critical Path in a network flow diagram, for the first node(event), we start with time ----

Select the correct option

- with strict positive value
- t = 0
- t = a (arbitrary)
- t = infinity



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Quiz Start Time: 09:00 AM, 29

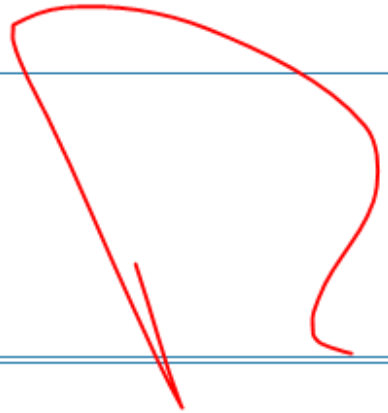
Question # 27 of 30 (Start time: 09:07:36 AM, 29 June 2021)

To

Solution region of the constraints: 2x+3y>12 or 6x+9y=36, will be the half plane bisected by '2x+3y=12', ---------- all the points on '2x+3y=12'.

Select the correct option

- excluding
- including



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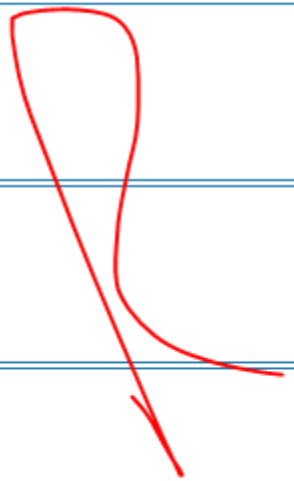
Question # 28 of 30 (Start time: 09:07:50 AM, 29 June 2021)

Which of the following event has no immediate predecessor in any network flow diagram?

Select the correct option

- ○ Critical Event
- ● Head Event
- ○ Tail Event
- ○ Non-Critical Event



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Question # 29 of 30 (Start time: 09:08:06 AM, 29 June 2021)

Which one of the following is not an operations research problem solving steps?

Select the correct option

- variable selections
- definition of the problems
- observation
- model construction

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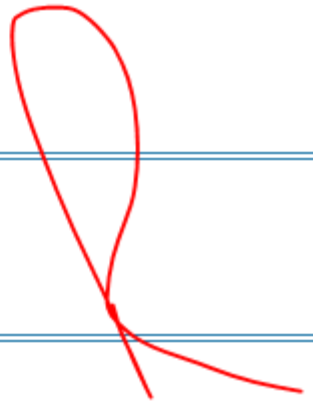
Question # 30 of 30 (Start time: 09:08:23 AM, 29 June 2021)

For critical path of a network, which of the followings is the best suitable answer.

Select the correct option

- ○ Its total float is zero
- ○ Its free float is zero
- ● All of the above
- ○ Its independent float is zero



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